LESSON 4: NATURE OF RESEARCH

Definitions, essential factors, characteristics, difficulties, the good researcher

PURPOSIVE COMMUNICATION | FINALS | DR. ARJAY B. ARCENA

What is research?

From the Old French **cerchier**, "to **seek or search**". Research is not a one-time search but **re-searching**: studies build on earlier studies (one pioneering COVID-19 study in 2020, then many quanti, quali and mixed-method ones). Research may be **replicated, never duplicated**.

DEFINITIONS (WHO SAID WHAT)

- **Thomas and Nelson (1996)** – "a **careful and systematic** means of **solving problems**"
  - **Careful** = don't **jump to conclusions**; studying only to get the result you want is not research (not **objective**)
  - **Systematic** = follows the **system (methods)**: problem → objectives/SOP → instrument → methods → results → summary → conclusion → recommendation; you can't skip a step
- **Merton (n.d.)** – research contributes to the **development of theory**: it **clarifies** concepts, **initiates**, **reformulates** and **refocuses** theory
- **Wiersma (1995)** – a **process**, an activity in the search for knowledge through **recognized methods of data collection, analysis and interpretation**
  - Needs detailed data-collection methods, including the **Data Privacy Act (RA 10173)**: collect only the data you need
- **Good (n.d.)** – a **systematic, defensible pursuit of results** that will **solve a problem**; good if it is an **ordered, reasonable, communicable** search
  - Ordered = **systematic**; reasonable = **objective**; communicable = **can be interpreted** (Very High, High, Low…)
- **Aquino (n.d.)** – the **systematic search for pertinent information** on a specific topic or problem; after analyzing and interpreting the data, the researcher faces another task: **preparing the research report**
  - Finding the answer is not the end: plan to **present or publish** it (say so in your methods)

Note: Memory tip: "**T**homas **S**olves, **M**erton builds **T**heory, **W**iersma has **M**ethods, **G**ood is **O-R-C** (ordered, reasonable, communicable), **A**quino writes the **R**eport."

ESSENTIAL FACTORS FOR EFFECTIVE RESEARCH: THE 3 MS

- **Money** – printing, paying advisers, panelists, validators, data analysts, debriefers, tokens
- **Materials** – paper, computer, printer, and the **instruments** you use
- **Men** – the **researchers**, **advisers**, **panelists**, **experts** (validators, analysts)

CHARACTERISTICS OF A GOOD RESEARCH

DE LEON AND CLAUDIO (4: "PRODUCES… RESULT")

- **Processible** – can be **discussed and interpreted**
- **Specific** – every question has its **own specific result** (SOP 1.a–1.e → Tables 1.a–1.e)
- **Generalizable** – lets you **generalize, conclude, recommend**
- **Interpretable** – the results can be interpreted

BEST (7)

- **Gathers new data** – not a duplicate
- **Solves problems**
- **Develops sound theories**
- **Requires expertise** – hence the experts (the "men")
- **Involves accurate observation and description** – only what you **really found**
- **Is logical and objective** – parts connected; **not biased**
- **Is carefully recorded and reported** – presented

THOMAS AND NELSON (1996) (5)

- **Systematic** – follows the system/procedure
- **Logical** – parts connected to one another
- **Empirical** – gives **evidence and support**
- **Reductive** – can be **reduced to its simplest interpretation** (simplified, explained)

- **Replicable** – same variables with another method, or another variable with the same method; **not duplicable**

Note: Memory tip: Thomas and Nelson = "**S-L-E-R-R**" (Systematic, Logical, Empirical, Reductive, Replicable): "**S**he **L**eft **E**arly, **R**an, **R**eturned." Best = 7, Leedy = 7, de Leon and Claudio = 4 (all start with "**Produces**").

PAUL LEEDY (7)

- **Begins with a question** in the mind of the researcher ("What is the problem?")
- **Requires a plan** ("How will I do it?")
- **Demands a clear statement of the problem**
- **Deals with the main problem through subproblems** – the A, B, C… of an SOP; probing questions in qualitative
- **Seeks direction through appropriate hypotheses** – quantitative designs
- **Deals with facts and their meaning**
- **Is circular** – the **recommendation** becomes a **new problem** for another study

Note: Look-alikes: "**logical and objective**" is **Best**; "**logical**" alone (with systematic, empirical…) is **Thomas and Nelson**. "**Systematic**" in a definition = **Thomas and Nelson**, **Good** or **Aquino**; as a characteristic list = **Thomas and Nelson**.

DIFFICULTIES ENCOUNTERED IN RESEARCH

- **Difficulty of recognizing problems** – find them in **literature**, past studies' **recommendations, scope and limitations, methods**, or by **observing** around you
- **Duplication** – copying a study that was already done (**replication** is acceptable)
- **Indifference of people to research** – not recognizing who the **right participants** are → use **inclusion and exclusion criteria**

A GOOD RESEARCHER (CALMORIN, 2004): R-E-S-E-A-R-C-H-E-R

- **R – Research-oriented** – has a background in research
- **E – Efficient** – revises **right away** (the day after the defense, while the minutes are fresh)
- **S – Scientific** – facts and observation; **cross-check** AI answers; **Wikipedia is not reliable** (anyone can edit it)
- **E – Effective** – delivers what the panel looks for; changes what needs changing
- **A – Active** – does the tasks asked (e.g. finds **replacement participants** when some back out)
- **R – Resourceful** – finds a way (email other participants; reuse paper; ask to submit by email)
- **C – Creative** – in **writing**: varied sentences (simple, compound, complex, compound-complex), **no run-ons**, restated not repeated (not Sailor Moon covers)
- **H – Honest** – objective; report only what was found
- **E – Economical** – saves money (online interviews recorded by GMeet/Zoom; ₱50 data instead of fare and food) but still gives **useful tokens** (pens for students, a handkerchief for mothers)
- **R – Religious** – pray, but above all **honest**: right participants, right number, results **not altered** to match what you wanted

END OF LESSON

PROPOSAL GUIDE: DR. ARCENA'S TIPS

For the prefinal defense and the final paper

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What do we submit?

No finals exam. Prefinal: defend your **proposal individually** to him. **Finals**: the **final paper**. The study continues in the **2nd semester** (Writing Across the Disciplines). It is **output-based**.

THE SYSTEM (EVERY PART MATCHES)

- **Read literature → find the problem and gap → choose the theory → objectives → SOP/research questions → instrument → methods → results → discussion → summary → conclusion → recommendation**
- **Objectives** come from the **title, design and theory**; the **SOP restates the objectives** as questions
  - Quantitative: **statement of the problem** (question form) | Qualitative: **intent of the study** | publications: research questions/objectives
- **Parallelism:** 3 SOP questions → **3 tables → 3 discussion parts → 3 summary parts → 3 conclusions**; sub-questions 1.a, 1.b → Tables 1.a, 1.b
- Clear presentation = readers **understand easily**; a confusing layout defeats its purpose

TITLES

- **Spell out** every acronym the first time: "Accelerated Pathway for Medicine 1 Students of **Davao Medical School Foundation, Inc.**"; APMED and DMSF afterwards
- **Don't put the design** in the title: "lived experiences" already says phenomenology; "writing proficiency" already says quantitative-descriptive
- **Don't put your conclusion in it:** "**Negative** Effects of…" or "Negative Experiences of…" is **biased**; maybe students were happy there was no class during the quake
- **Be specific** (the S of **SMART**) and **Time-bounded** (the T): pick a study you can finish on time
- Easy studies are fine; impact matters more than complexity ("Yawa", the Tagalog "Ay", "The Many Mukha of Faces")

THEORY: FRAMEWORK OR LENS

- **Quantitative:** a theory is **required**; it is the **anchor** and gives the **indicators** in your SOP
  - Example: **Jacobs et al.**, writing proficiency = **VOCFM**: **V**ocabulary (incl. grammar), **O**rganization (intro–body–conclusion, connectors), **C**ontent (strength of support), **F**luency (smooth, exact words), **M**echanics (punctuation, run-ons, fragments, margins)
  - Same theory name, different indicators across authors: **choose carefully and keep your terms consistent**
- **Qualitative: theoretical lens**, often **left out** when proposing if you don't yet know what you'll find (lived experiences, a ritual you haven't observed); you may use **several** theories later
  - Use one when you already know what you're looking for (**politeness markers → Politeness Theory**)
  - **Grounded theory** uses **no** theory: you **build** one
- Hypothesis (tested) → **theory** (still questionable, can be refocused) → **law** (no longer questioned); "I have a feeling…" is a hypothesis, not a theory

OBJECTIVES → SOP → HYPOTHESIS ("TRANSLATING")

- Objective: "To **identify the level** of writing proficiency of APMED first-year students in terms of vocabulary, organization, content, fluency and mechanics."
- SOP: "**What is the level** of writing proficiency of APMED first-year students in terms of: a. vocabulary… e. mechanics?" then "What is the **overall** level…?" (an optional first question: **demographic profile** by sex and school last attended)
- Hypotheses only for designs that look for **significance**:

- **Comparative** – significant **difference** | **Correlational** – significant **relationship** | **Causal** – significant **influence** | **(Quasi-)experimental** – significant **effect**
- **Ho:** "There is **no** significant…" | **Ha:** "There **is** a significant…"

- **No hypothesis:** single-variable **descriptive** studies ("single ka, wala kay karelasyon") and **all qualitative** studies (no variables to measure); qualitative may state an **assumption** instead

INSTRUMENTS AND ANALYSIS

- Fit the instrument to the variable: writing proficiency → an **essay prompt** (200+ words, IBC structure, format rules), checked by **experts** (English teachers); a self-rating survey only shows **perception**, not proficiency
- **Likert scale: 5-point** (Very Low → Very High) or **4-point** (no neutral middle, so people can't just pick "3")
  - Three columns: **mean**, **description**, **interpretation**; the **interpretation** is what you use in the **conclusion** (summary = mean + description)
- **Statistical treatment:** demographics → **frequency** (just counting); levels → **mean and standard deviation**; no t-test when you compare nothing
- **Design:** say exactly which: **quantitative-descriptive** (not just "descriptive"); qualitative "general descriptive" when it isn't pheno, ethno, case, narrative or grounded theory
- **Data Privacy Act (RA 10173):** ask only for needed data (no name, section, program if unused)
- Researcher-made instruments: statements must come from your **literature**; that is their basis and what supports your results

QUALITATIVE REMINDERS

- **No conclusion** in qualitative (5–20 participants per Creswell; 1 per Dukes; up to 325 per Polkinghorne; usually **convenience** sampling, so they can't represent everyone) → write **implications**
- Quantitative can conclude (e.g. a **simple random sample** of 30% represents the class)
- Reduce bias: **probing questions** ("Did I understand correctly that…?"), **member checking** (Braun and Clarke's last step: go back to participants), a **debriefer** to check your thematic analysis

ETHICS WITH THE PEOPLE WHO HELP YOU

- **Validators:** give at least **3 days**; pay or give fair snacks unless they say it's free; **don't serve food you wouldn't eat** or cheaper than your own (his "ginanggang vs burger" story); thank them in the **acknowledgment**
- **Panelists** check **technicalities** (citations, connected parts), not just margins; **advisers** guide what and how
- **AI:** fine for ideas if **the idea is yours** and you prompt it; **don't just copy-paste**; **cross-check** references
- **Present or publish** your paper (fora, conferences) so the work isn't wasted

END OF LESSON